

The Dwarf Pilot: four experts, two rounds, one verdict

Can memory-write behavior identify a Berkeley numerical dwarf? A DSP engineer, an ML engineer, an ECG signal scientist, and a volatile-memory systems engineer reasoned in four phases — alone, together, alone again, together — over the write-visible kernel corpus. This is each expert's final standalone verdict and the joint one.

Method: independent Phase-1 verdicts → Phase-2 cross-examination → Phase-3 revised solo verdicts (below) → Phase-4 synthesis. All claims grounded in kernel source, the differ metrics, and the capture config; estimates labelled. No capture has run yet — every separability statement is a falsifiable prediction, not a measurement.

§0 The one-line result

"Dwarf" is not a learnable target. The 500 ms snapshot is a 2 Hz stroboscope that integrates thousands of compute passes into "footprint fully lit," collapsing the 13 dwarfs into a **~6–8-cell region of a two-axis space**: an always-visible *spatial image* (footprint size × saturation × per-page magnitude) and a *conditionally-visible temporal trajectory* (front migration / magnitude decay) that survives only for a handful of high-arithmetic-intensity kernels once deliberately resized. The founding byte-domain FFT lens is demoted. A single ~1–2 hour gated pilot decides whether this earns a full campaign or whether the honest deliverable is the dwarf→archetype confusion matrix the spec asked for.

§1 Joint verdict PHASE 4

1 • The target is a product space, not a list

Two of the four experts (ML and ECG) arrived *independently* at the same decomposition. The dwarf is a region of a two-axis product; the honest ceiling is ~6–8 populated cells for the numerical dwarfs, not 13.

AXIS 1 — PER-APSHOT IMAGE

ALWAYS OBSERVABLE

Footprint **size** × **saturation** (dense-full vs sparse-stable) × per-page **magnitude regime** (high-entropy re-seed vs low-amplitude smooth). This is the ~3–5 write-shape archetypes — the instantaneous shape.

AXIS 2 — ACROSS-APSHOT TRAJECTORY

CONDITIONAL

Time-series shape of footprint extent/centroid and magnitude envelope: **static / grow / shrink / translating-band / decaying**. Observable only for high-intensity fronts sized up, plus converging iteratives.

Deliverable framing: predict the *cell*, and report the **dwarf→cell confusion matrix** — which is what `test_families_spec.md` set out to produce in the first place.

2 · Temporal — resolved per-kernel by an arithmetic-intensity law under a hard RAM cap

Rhythm is dead (2 Hz Nyquist). The one-time *evolution envelope* (LU's shrinking front, QR's growing front, a solver's convergence) is a sub-Hz trend, not a rhythm, and is rescuable — but **only for kernels whose work grows faster than their footprint** (arithmetic intensity = flops per byte written). The 1 GiB guest RAM is the binding constraint.

Kernel	Intensity	Default N	Target N (~4s/pass)	Footprint	Guest RAM	Verdict
QR	$O(N^3)/O(N^2)$	1024	~2000 (2×)	~7,600 pg	~3%	Cheap rescue — first
LU	$O(N^3)/O(N^2)$	1024	~3,800 (3.7×)	~27,000 pg	~11%	Moderate — second
DP (edit-dist)	$O(N^2)/O(N^2)$	8192	~58,000 (7×)	13.4 GB	1340%	Infeasible
Stencil (Jacobi)	$O(1)$ flops/byte	1024	~44,400 (43×)	31.6 GB	3080%	Infeasible + moot
FFT (1D)	$O(N \log N)$	2^{20}	~ 2^{27}	2.1 GB	210%	Infeasible

Rescuable temporal arm: the dense $O(N^3)/O(N^2)$ fronts (QR, LU, and by the same law Cholesky, Floyd, Smith–Waterman). **Permanently aliased:** the $O(1)$ -flops/byte saturators (stencils, multigrid, nbody, LBM, FDTD, DWT) and the memory-bound DP/FFT — they cannot be slowed past a snapshot without overflowing the guest. These reclassify into a *footprint-size, no-front* bucket; their only signal is fixed footprint magnitude and, for iteratives, magnitude decay.

Caveat (memory): the QR/LU targets are extrapolated from cache-resident smoke timing. At $N \approx 2000-4000$ the working set blows past L2/L3, so real per-pass time scales *worse* than N^3 — meaning the true N is likely *smaller*, and the rescue is *stronger*. Gate 1 (a real timing readout) must confirm before capture budget is spent.

3 • The features — a placement-invariant core, a gated address-shape extension

- **Demoted:** the founding byte-domain complex-page FFT / wavelet / cepstrum lens — aliased by the stroboscope and switched off at `substrateSpeed 2`. ECG withdrew its own multigrid-cepstrum; cepstrum is even the wrong tool for a monotone ramp (no quefrency peak — use ramp slope/curvature).
- **Surviving leads (all four agree):**
 1. **Per-page magnitude distribution** (hamming / l1 / l2) — separates smooth low-amplitude writers (n-body, barnes-hut, MD, gibbs, belief-prop) from high-entropy re-seeders. The founding intuition's real payoff, and exactly what APF's binary changed/not indicator destroys.
 2. **Footprint trajectory** (changed-page-index extent/centroid series) — for resized high-intensity fronts only.
 3. **Magnitude-envelope decay** — converging iteratives (gibbs/BP) whose later overwrites flip fewer bits.
- **Model:** no autoencoder bank (degenerate on near-flat inputs in B1; a forest beat it 19–59× on identical inputs). Discriminative trees on a small feature vector. **Effective n = dwarfs, not windows** — a trajectory is one realization per cell, so a generalization campaign needs ~100–135 cells.
- **Memory's schema catch:** the existing `change_centroid/span/runs` channels are *per-page* (byte offsets inside one 4 KiB page), *not* whole-dump position. The address-space shape feature must be built from the `page_index` column — and that is the feature most exposed to guest-physical scrambling. Placement-invariant aggregates (run-length distribution, total footprint size, magnitude histograms) are the safe core.

4 • Guest-physical contiguity — the deepest validity threat, and it is free to test

Kernels get virtual contiguity from one `mmap()`, but `pmemsave` dumps guest-*physical* RAM, and the guest buddy allocator decides the mapping. All 61 phase-2 kernels call `MADV_NOHUGEPAGE` (a deliberate THP opt-out), which *removes* the one mechanism that would force physical contiguity — so the concern is more exposed, not less. But the differ already emits `page_index`, so the test costs nothing extra: capture `gemm` at default for ~15 s with `--speed 0 --sparse`, take two consecutive snapshots deep in the measure phase, and check whether the changed pages form a contiguous run or scatter across the 262,144-page range. This single check gates every address-shape claim in the pilot.

5 · Leakage guards (B1's hollow success, sharpened for trajectories)

- **Split unit = DWARF** for the cross-motif claim (B1's LORO leaked: 0.836 vs 0.457). Never cell/rep.
- **IDLE controls are the tripwire:** `spmv/bfs/mc_pi` must stay mutually inseparable at floor after masking. If they separate, you are reading the environment, not the workload.
- **Level-normalization:** separation must survive removing absolute write volume — else you rebuilt APF.
- **Trajectory-specific:** normalize to *trace-fraction* (kills the duration confound) and **drop warm-up snapshots** (the allocation/page-fault-in transient is a per-workload fingerprint).
- **Normalize the capture protocol by pass-duration, not by dimension** — so footprint size cannot leak as the dwarf label.

6 · The pilot — unanimous, gated cheapest-first

Gate 0 — arithmetic, done, zero cost. The intensity table above. Output: QR→~2000, LU→~3800 as resize targets; DP/stencil/FFT ruled out for resizing.

Gate 1 — server, seconds per kernel, no capture pipeline. Run `kernel_qr_v2 --dim 2000` and `kernel_lu_v2 --dim 3800` standalone (`--dry-run` first), read the factorisation counter. Confirms the cache-effect caveat before any capture budget is spent.

Gate 2 — first real capture, small, decisive. `gemm` default ~15 s at `--speed 0 --sparse` → the contiguity test. **The single most important go/no-go in the whole pilot** — every address-shape feature depends on it.

Gate 3 — pilot proper, `--speed 0`, duration ≥70 s (145 frames, clears the 128-frame streaming threshold), only if Gate 2 passes:

- resized **QR + LU** — front migration via Jaccard overlap of consecutive changed-page-index sets (overlap <1 with a shifting boundary = front survived; overlap ≈1 = still aliased);
- **gemm** default — WORKING-SET / static null;
- **barnes_hut vs nbody** — the address-shape pair, most sensitive to Gate 2;
- **gibbs** — thinnest visible control (256 pages), magnitude-decay envelope;
- **stencil_jacobi** default — ECG's *falsification control*: the law predicts it stays saturated and static; if it separates, the intensity law is wrong;
- **dp** reduced to `--dim 2048` (not the coded 8192 — backpressure risk);
- ML's cross-motif extension: add **rmat + lexer** (also spec-SEQUENTIAL-GROW) so QR can be held out *leave-one-dwarf-out* — train on rmat+lexer, does held-out QR read “growing”?

GO Resized LU/QR (and the growing triad) separate from GEMM on the trajectory lead, *and* barnes_hut/gibbs separate on the magnitude lead, *and* the separation survives level-normalization, placement-invariant-only features, and warm-up dropping → the multi-lens (minus byte-FFT) is vindicated. Commit the ~100-cell campaign at controlled pass-duration.

NO-GO Resized LU/QR still collapse into GEMM → the front was workload memorization, temporal is dead even sized (ML's position wins outright), and Part 2 honestly reduces to a spatial magnitude + footprint-size classifier. Deliver the **dwarf→archetype confusion matrix** and stop spending server time on the temporal arm.

7 • The one residual disagreement — pilot speed

ECG voted **speed 2** (the surviving leads are alive there, and it matches production cadence). Memory voted **speed 0** (the rich magnitude channels, the contiguity test, and a free byte-FFT A/B all need it; speed 2 is a documented throughput compromise).

Resolution: run the *pilot* at speed 0 — cheap at pilot scale (only a few thousand changed pages), and it guarantees axis C, contiguity, and the byte-FFT A/B are all observable at once. Then confirm axis C survives at speed 2 before the campaign runs at production cadence. DSP's "free speed-0 A/B on the same retained dumps" is the bridge.

§2 The four solo verdicts PHASE 3

Each expert's final standalone position after the cross-examination. Condensed from their own words; unverified items are flagged as they flagged them.

DSP engineer — the sampling lens

Verdict. The governing filter is the 500 ms boxcar-OR, a 2 Hz stroboscope. At default sizes it aliases every kernel's intra-pass geometry to "footprint fully lit," so the visible dwarfs collapse into ~5 write-shape archetypes = a small grid of (footprint-size × magnitude) buckets. Rhythm is dead; ML is right on that. But the *evolution envelope*

(LU shrink, QR grow) is a DC-to-0.05 Hz trend, not a rhythm, and is rescuable — the key arithmetic: to slow one evolution to span many snapshots while fitting 1 GiB, **work must grow faster than space**. Dense LA ($O(N^3)/O(N^2)$) qualifies; append-growers and band-migrators (work $\approx$ space) fold.

What he conceded/withdrew. Adopted ML's placement-invariance as a hard filter and *withdrew his own "contiguous-vs-scatter shape" feature* — ASLR + guest-physical fragmentation make absolute position non-reproducible. What survives: translation-invariant stats (footprint size, run-length histogram) and temporal derivatives of spatial moments (centroid velocity, span growth-rate), which are invariant to the random origin. Corrected ECG: read convergence-decay on `l1/l2`, not `hamming` (a 1e-10 update still flips low mantissa bits, so hamming won't decay); and a converging solver shows footprint shrinkage *and* magnitude decay together, which distinguishes it from an LU shrink at constant front magnitude — the joint (size(t), l2(t)) trajectory is the discriminator.

Pilot vote. Two arms, both speed 2 (+ a free speed-0 A/B on retained dumps): Arm A the collapse null (GEMM, stencil, hmm, diffusion — if they don't separate, archetype is the ceiling, deliver the confusion matrix); Arm B the evolution rescue (GEMM/LU/QR resized + a slowed stencil — if the {static, shrink, grow, decay} trajectory classes separate, the sub-type is real).

ML engineer — the representation / leakage lens

Verdict. Neither "family" nor "dwarf" is a coherent learnable target. The identifiable target is a point in a three-axis product: A = instantaneous footprint density (~ 3 levels at 2 Hz; SCATTER largely collapses into dense under the boxcar-OR), B = footprint trajectory, C = per-page magnitude distribution and its evolution. Dwarf is recoverable as a *region* (~ 8 – 15 cells) — dwarf-informative for the numerical dwarfs but not reaching 13 (Jacobi/Floyd/Gibbs share a cell).

The catch he owns. Axes B and C are **once-per-cell trajectory features**: one LU run = one shrinking curve, and the windows inside it are points on that curve, not independent samples. So effective n collapses from thousands of windows to one realization per cell, and a defensible cross-motif claim needs ≥ 3 dwarfs $\times \geq 3$ workloads $\times \geq 3$ reps ≈ 100 – 135 cells. The target got richer and the data got thinner at the same time — that tension, not signal presence, is what the program lives or dies on.

Two new leak guards: normalize each trajectory to trace-fraction (duration confound); drop warm-up snapshots (the allocation transient is a workload fingerprint — a

trajectory-flavored rerun of B1's floor recognition). **Pilot vote:** a sized, gated, growing-front *generalization triad* — QR, rmat, lexer (three different motifs the spec all calls SEQUENTIAL-GROW) + GEMM contrast + spmv idle-null, speed 2, leave-one-dwarf-out on the axis-B label. Held-out growing dwarf recognized across motif (surviving the three guards) → GO; collapses → archetype is the ceiling, deliver the confusion matrix. Unverified and gating: whether speed 2 actually emits per-page magnitude, or only the binary indicator — if not, axis C needs speed 0.

ECG signal scientist — the morphology / analogy lens

Verdict. The write-visible dwarfs do not carry 13 separable morphologies. After the stroboscope integrates thousands of sub-snapshot passes (his own gate: 16–30,000 passes/snapshot at smoke sizes), the target collapses to a small set of cells on two axes — an always-present spatial image and a conditionally-present temporal trajectory. The founding byte-domain FFT/wavelet/cepstrum lens is demoted.

His law. The temporal rescue is governed by arithmetic intensity (flops per byte). High-intensity fronts ($O(N^3)/O(N^2)$: LU, QR, DP, Floyd, Cholesky, Smith-Waterman) can be sized so one pass spans ≥ 3 snapshots under the cap; low-intensity saturators (stencils, multigrid, nbody, all $O(1)$ flops/byte) never can, at any feasible size. He *withdrew his own Phase-1 multigrid-cepstrum* as impractical and conceded multigrid to ML. Cepstrum is the wrong tool for a monotone ramp anyway — use the change-centroid ramp slope/curvature; reserve cepstrum for the one surviving recurrence, the convergence-decay envelope, which he adopts.

Falsifiable prediction + pilot. Only the high-intensity fronts separate from GEMM on the footprint-trajectory lead; low-intensity kernels stay inseparable except by footprint size, at every feasible size. Pilot adds a *negative control*: capture a maximally-resized stencil alongside — the law predicts it stays aliased; if it separates, the intensity rule is falsified. Speed 2. Flagged a discrepancy: in his smoke read, qr/lu/dp emit no pass counter and stencil is 30,157/snap, so memory's "QR ~0.9" is from a larger sizing — which is exactly the resizability curve, and what Gate 1 resolves.

Volatile-memory systems engineer — the ground-truth lens

Verdict. Aliasing is a per-kernel continuum spanning four orders of magnitude at the same cadence, which is what makes sizing-rescue a targeted question rather than a blanket “temporal is dead.” None of these dwarfs has a *volume* problem (they write 2×–555× the ~118-page OS floor); ~15 of ~20 collapse into the same footprint-magnitude bucket regardless of identity, which floor-subtraction does nothing to fix. Speed 2 dropping the heavy channels is documented and deliberate (`--help` : use speed 2–4 at 500 ms) — the magnitude/texture leads require an explicit `--speed 0` .

His hard numbers. The per-kernel resize table (QR ~2000 at ~3% RAM; LU ~3800 at ~11%; DP 13.4 GB, stencil 31.6 GB — both infeasible on a 1 GiB guest) turns ECG's law into RAM-bounded fact. The cache-effect caveat makes QR/LU's real N smaller, strengthening the rescue — but Gate 1 must confirm. Two source-read mislabel root causes: `sddmm` 's sparsity is index-metadata invisible to a byte differ (inherent limit), while `fem_assembly` zeroes the whole matrix every pass (a fixable coding choice).

The contiguity gate. All 61 kernels opt out of THP (`MADV_NOHUGEPAGE`), so physical contiguity is more exposed — but the differ already emits `page_index` , so a single ~15 s gemm capture at `--speed 0 --sparse` settles it before any address-shape feature is trusted. **Vote:** proceed, gates cheapest-first (0 arithmetic → 1 server timing → 2 contiguity → 3 pilot at speed 0, ≥70 s). If Gate 2 shows severe scattering, drop barnes_hut/nbody and the LU/QR-front claims, keep only magnitude + footprint-size, and treat contiguity as a separate follow-up.

Four-expert deliberation, memory-signal thesis Part 2. Phases 1–4. Design only — no capture has run; all separability statements are predictions to be tested by the pilot above. Kernel sources:

`VM_executables_phase2/kernel/` ; differ metrics:

`VM_sampler/VM_Capture/live_delta_calc_modular/src/metrics/` ; capture config:

`config_qemu_upc.json` (500 ms, 1 GiB); taxonomy: `docs/test_families_spec.md` .